RESPONSE PAPER

While the two presented papers focus on distinct technological domains, they converge on a critical theme: the necessity of structured data management, transparency, and standardization in the face of rapidly evolving digital landscapes. The paper "Datasheets for Datasets" addresses the documentation of the lifecycles of datasets that underpin machine learning (ML) models. In contrast, the study "Federated Data Modeling for Built Environment Digital Twins" seeks to ensure data interoperability within the architecture, engineering, construction, and operations (AECO) sector. Both studies demonstrate that systemic shortcomings in current data practices directly hinder the reliability, accountability, and large-scale integration of modern digital infrastructures.

In "Datasheets for Datasets," the authors highlight a significant lack of standardized documentation regarding the creation and deployment of datasets within the machine learning community. This transparency gap can lead to severe consequences in high-stakes domains such as criminal justice, hiring, and finance where algorithms may inadvertently reproduce or amplify societal biases present in the training data. To mitigate these risks, the authors propose a "datasheet" standard inspired by the electronics industry. This protocol mandates that every dataset be accompanied by documentation detailing its motivation, composition, collection process, and recommended uses, thereby increasing transparency and facilitating the reproducibility of results.

Conversely, Moretti et al. address the fragmented and heterogeneous nature of data within the AECO industry. While smart building management requires the integration of real-time IoT sensor data with static geometric and contextual information, these assets are often trapped in closed ecosystems or proprietary formats, stifling interoperability. Rather than forcing all data into a single, massive integrated structure, the authors propose a "federated" data modeling approach. By interconnecting existing open models such as IFC, BrickSchema, and Crate through core interface entities, data can remain authentic and manageable within its specific domain without redundant duplication.

In conclusion, these texts collectively illustrate that the value of modern digital infrastructures is derived not merely from the act of "collecting data," but from the quality, traceability, and interoperability of those data assets. Just as meticulous documentation is fundamental to developing ethical and fair algorithms, a federated structure is vital for enabling sustainable decision-making within digital twin ecosystems. Both papers urge technology developers to move beyond viewing data as a raw material, treating it instead as a strategic asset that must be managed transparently and responsibly throughout its entire lifecycle.

Ali Kemal Akbulut